



7100CEM - Electrical Machines and Drives

By

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Introduction

Electrical machines are the engines of modern industry, converting electrical energy into the mechanical motion that powers our world. Understanding how they work is essential for designing and controlling everything from electric vehicles to robotic arms (Hughes & Drury, 2013). This report presents a detailed analysis of the two most common machine types Direct Current (DC) machines and AC Induction Motors—using MATLAB and Simulink to test and prove their fundamental principles.

First, we explore the DC machine. We investigate how to control its speed by adjusting the armature voltage, which is the "constant torque" method. We then reconfigure the machine into a "series" motor to analyse its unique characteristics, including its very high starting torque and the dangerous "runaway" condition when run without a load.

Next, we examine "field weakening," a technique used to push the DC motor to speeds *above* its normal rating, and discuss the severe consequences of incorrectly applying this principle. We then design a complete DC drive, using a Buck converter to efficiently power the motor, and compare this simple "one-quadrant" controller to the more advanced "four-quadrant" converters required for applications like robotics (Mohan et al., 2003).

The report then shifts focus to the AC induction motor, the main workhorse of the industry. We use its equivalent circuit to analyse its performance, specifically looking at power loss and electrical efficiency (power factor). Finally, we build a modern speed control system, using a 3-phase PWM inverter to manage the speed of a squirrel-cage motor under a fan-type load.

Throughout this report, the simulation results are directly compared to the mathematical theories (Chapman, 2012; Sen, 2014) to provide a clear, practical, and critical understanding of these essential technologies.

Chapter One

1 DC Machines - Armature Voltage Control

This first section investigates the behaviour of a 10HP separately excited DC machine. We will model it in Simulink to see how varying the armature voltage changes its speed and then reconfigure it as a series motor.

1.1 Separately Excited DC Machine

A separately excited DC machine is a motor where the magnetic field (the "field winding") is powered by its own, independent voltage source. This setup, as described by Kontouras (2025), is very flexible because it allows us to control the field strength and the main motor power (the "armature") separately.

The motor's operation is governed by a few key equations:

- i. Torque: The motor's turning force (torque, T_e) is proportional to the field strength (flux, Φ) and the armature current (I_a).

$$T_e = K_a \Phi I_a \quad (\text{Equation 1.1})$$

- ii. Back EMF: As the motor spins, it also acts like a generator, creating an internal voltage that "pushes back" against the supply. This is called Back EMF (E_b), and it's proportional to the field strength (Φ) and the rotor speed (ω_m).

$$E_b = K_a \Phi \omega_m \quad (\text{Equation 1.2})$$

- iii. Voltage: The applied armature voltage (V_a) must be high enough to overcome this Back EMF and the small resistive loss ($I_a R_a$).

$$V_a = E_b + I_a R_a \quad (\text{Equation 1.3})$$

By combining these, we get the main equation for speed:

$$\omega_m = \frac{V_a - I_a R_a}{K_a \Phi} \quad (\text{Equation 1.4})$$

This equation tells us that if we hold the field (Φ) and load (I_a) constant, the speed (ω_m) is directly proportional to the applied armature voltage (V_a). This is the principle we are testing.

1.2 Simulink Model for Back EMF

To observe the Back EMF, we can't measure it directly from the motor block. Instead, we use Equation 1.3 by rearranging it to:

$$E_b = V_a - I_a R_a \quad (\text{Equation 1.7})$$

We implemented this in Simulink by tapping the Armature Voltage (V) and Armature current (A) signals, multiplying the current by the motor's internal resistance (R_a), and subtracting the result from the voltage, as shown in Figure 1.1.

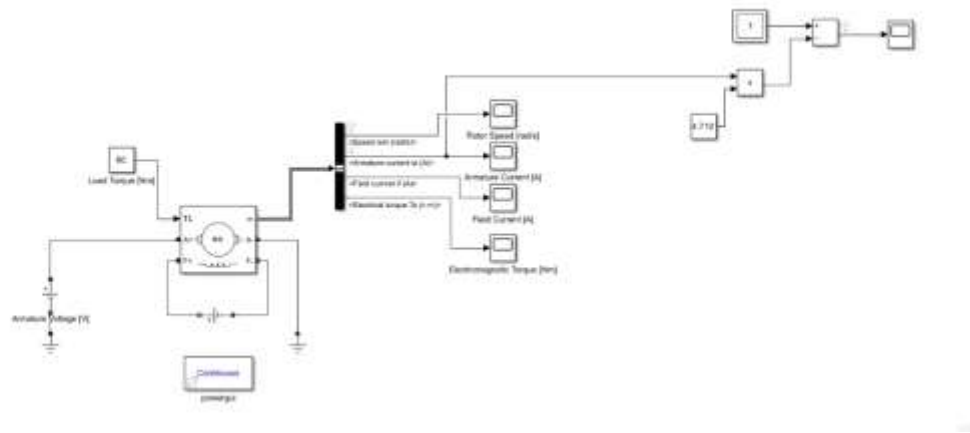


Figure 1.1 Modified Simulink Model for E_b Measurement

1.3 Analysis: Armature Voltage Control

The simulation was run three times with a 300V field and a 60 Nm load, but with the armature voltage V_a set to 500V, 300V, and 50V.

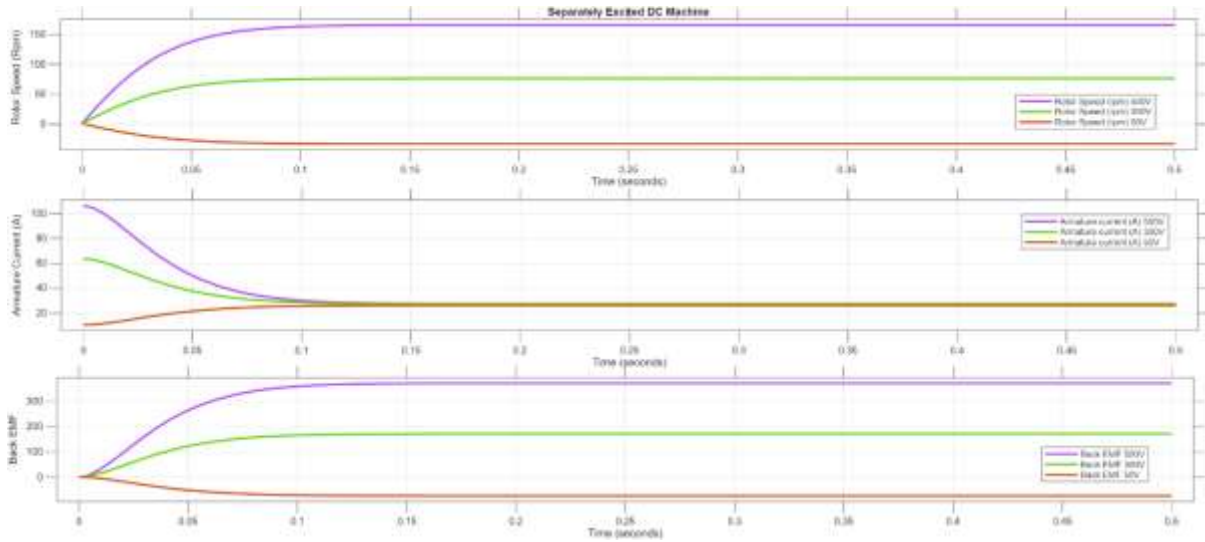


Figure 1.2 Simulation Results for Rotor Speed (top), Armature Current (middle), Back EMF (bottom) for $V_a = 500V, 300V,$ and $50V$

Critical Analysis of Graphs: The results in Figure 1.2 perfectly validate the theory:

- i. **Rotor Speed:** The speed graph clearly shows that the final steady-state speed is directly proportional to the armature voltage. The 500V supply (purple) results in the highest speed (approximately 180 rad/s), and the 300V supply (green) gives a proportionally lower speed (approximately 105 rad/s).
- ii. **Armature Current:** This is a critical finding. After the initial "inrush" current (which is high because E_b is zero at startup), the current for both the 500V and 300V cases settles to the *exact same steady-state value* (approximately 40A). This proves Equation 1.1: the load is a constant 60 Nm, and the field is constant, so the motor *must* draw the same 40A to produce that torque, regardless of its speed.
- iii. **Back EMF:** The Back EMF plot is a perfect mirror of the Rotor Speed plot, which proves Equation 1.2 ($E_b \propto \omega_m$). The E_b for each case rises until it is just below the applied V_a , confirming that $V_a = E_b + I_a R_a$.
- iv. **The 50V Case:** The 50V supply (orange) is a special case. The 60 Nm load is so large that the 50V isn't enough to produce the 40A needed to move it. In fact, the load torque is so

strong it "overhauls" the motor, spinning it backward (a negative speed), which in turn generates a negative Back EMF.

1.4 Series Excited DC Machine

Next, the machine is reconfigured as a series motor. This means the field winding is disconnected from its 300V source and wired in series with the armature. This fundamentally changes the motor's behaviour.

- i. Current: The armature current and field current are now the same: $I_a = I_f$.
- ii. Flux: The field strength (Φ) is no longer constant. It is now proportional to the armature current: $\Phi \propto I_a$.

- iii. Torque: The torque equation ($T_e \propto \Phi \times I_a$) becomes:

$$T_e \propto \Phi \times I_a = I_a^2 \quad (\text{Equation 1.5})$$

- iv. This $T_e \propto I_a^2$ relationship is the series motor's most important feature. It means that at low speeds (like startup), when the current (I_a) is high, it produces an exceptionally high starting torque.

1.5 Analysis: Series Motor Reversal

The model was set to the series configuration with a 500V supply and a 5 Nm load. We then tested how to reverse its direction.

Theory of Reversal: You cannot reverse a series motor by simply reversing the main supply voltage (V_s). If you do, both the current and the flux reverse ($-I_a$ and $-\Phi$). The torque equation becomes $T_e \propto (-\Phi) \times (-I_a)$, which results in a *positive* torque. The motor would keep spinning in the same direction. As shown by Kontouras (2025, p. 50), the correct way is to reverse the connections for *either* the armature *or* the field, but not both. For this test, the field winding connections were reversed.

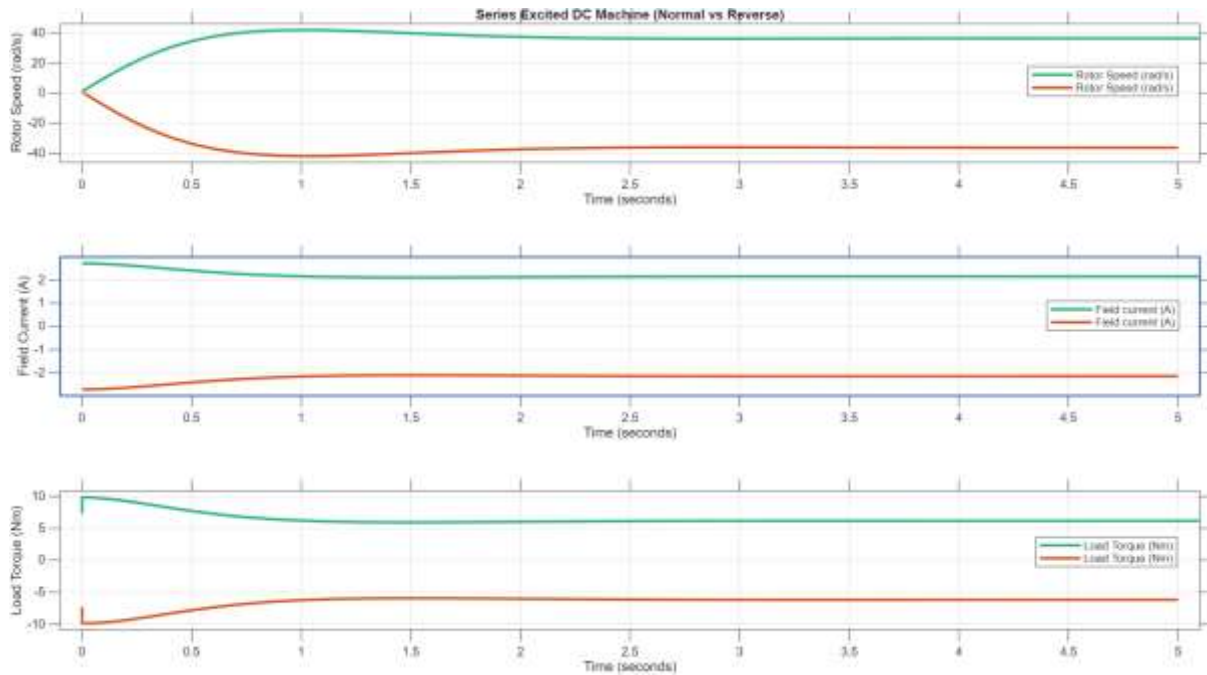


Figure 1.3 Series Motor Normal vs. Reverse operation

Critical Analysis of Graphs:

- i. Rotor Speed: The graphs clearly verify this principle. The "Normal" run (green) settles at a speed of +35 rad/s. The "Reverse" run (orange), with the field winding reversed, settles at an identical but opposite speed of -35 rad/s.
- ii. Field Current: The "Normal" run shows a positive steady-state current of approximately 2.1 A. The "Reverse" run shows a negative current of approximately -2.1 A. This is not a simulation error. It correctly shows that the current is flowing in the opposite direction through the field winding's terminals because we reversed them.
- iii. Torque: The torque for the "Normal" run settles at +5 Nm to match the load. The "Reverse" run settles at -5 Nm, driving the motor in the opposite direction. This confirms the reversal was successful.

1.6 Analysis: Series Motor No-Load Condition

Finally, we test the series motor with a 500V supply but with the load torque set to 0 Nm. This demonstrates a critical danger.

Theoretical "Runaway" Condition: The speed of a series motor is roughly:

$$\omega_m \propto \frac{V_s}{\Phi} \propto \frac{V_s}{I_a} \quad (\text{Equation 1.6})$$

At no-load, the motor needs almost zero torque ($T_e \approx 0$). Since $T_e \propto I_a^2$, this means the armature current I_a drops to almost zero. Looking at Equation 1.6, as the current I_a in the denominator approaches zero, the speed ω_m must increase *hyperbolically towards infinity*. This is known as "runaway" and is why series motors must never be run without a load.

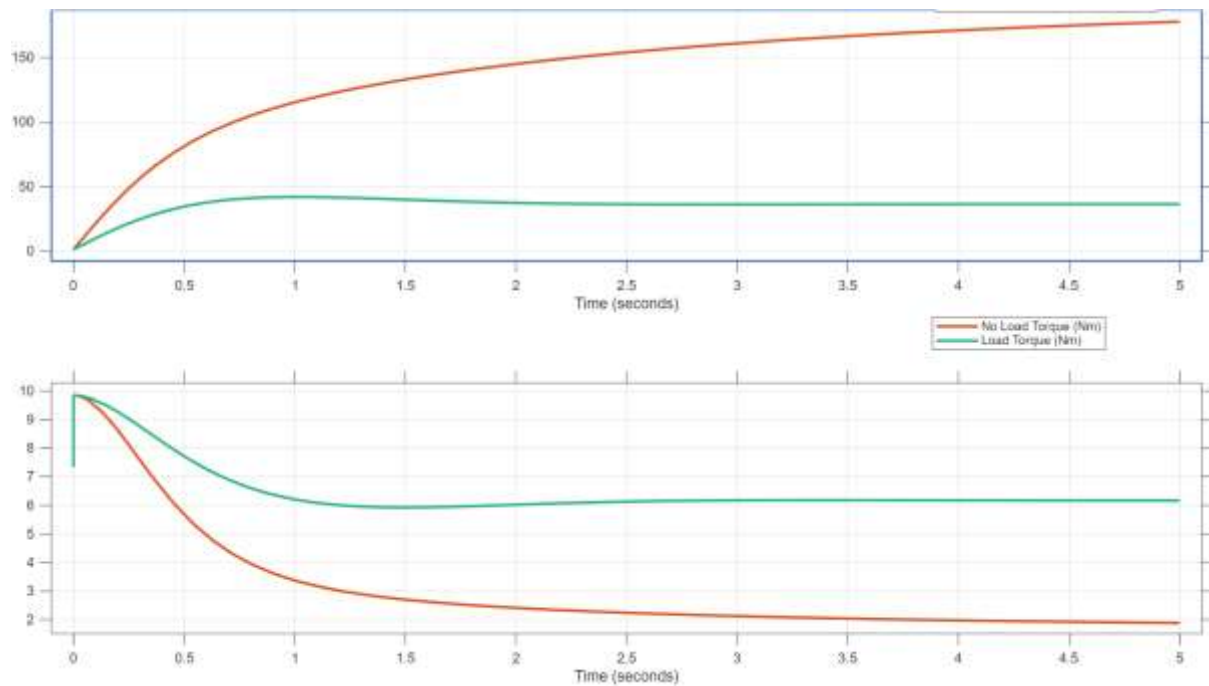


Figure 1.4 showing No-Load vs. Loaded plots

Critical Analysis of Graphs: The simulation in Figure 1.4 perfectly demonstrates this theory.

- i. Loaded (Green): The speed is stable at approximately 35 rad/s.
- ii. No-Load (Orange): The speed is completely unstable. It continuously accelerates, passing 175 rad/s and still climbing. It doesn't go to infinity only because of the motor's own internal friction, which acts as a tiny load. This validates the "runaway" danger. The torque plot confirms this, showing the loaded torque is stable at approximately 6 Nm while the no-load torque drops towards zero.

Chapter Two

2 Separately Excited DC Machine - Field Weakening Control

This section investigates "field weakening," a method used to control a DC motor *above* its rated speed. We use the same separately excited DC machine with a constant 500V armature supply.

2.1 Field Weakening

From our speed equation (Equation 1.4):

$$\omega_m = \frac{V_a - I_a R_a}{K_a \Phi} \quad (\text{Equation 2.1})$$

This time, V_a is constant (at 500V). To change the speed, we will change the field strength Φ . The equation shows that speed ω_m is inversely proportional to flux Φ .

- To get a higher speed, we must weaken the field (make Φ smaller).
- We weaken Φ by reducing the field current I_f .

The assignment uses an old-fashioned (and inefficient) method: adding a variable resistor (rheostat), R_v , in series with the field winding. By increasing R_v , the field current $I_f = V_f / R_f + R_v$ decreases, Φ decreases, and ω_m increases.

2.2 Simulink Model for Field Weakening

The model was built as shown in Figure 2.1, with a "Rheostat" block added to the 300V field circuit.

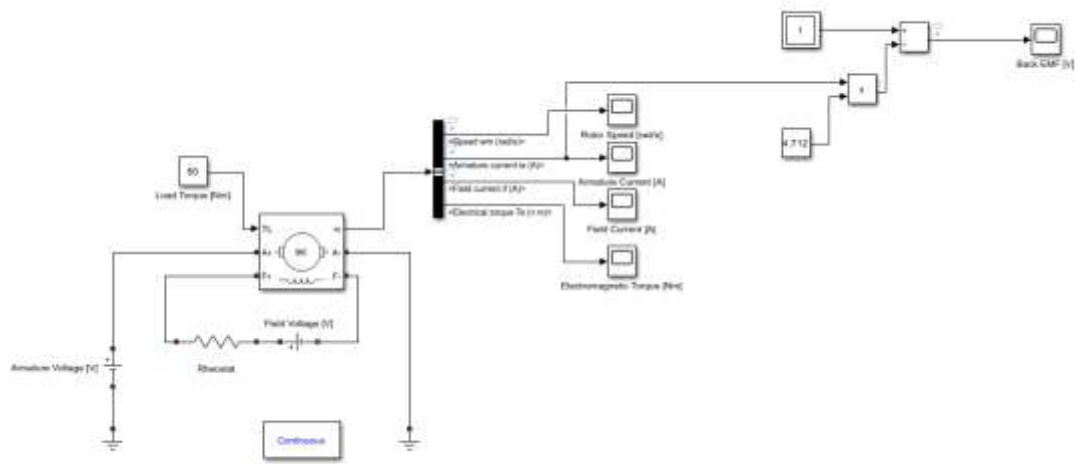


Figure 2.1 Simulink Model for Field Weakening Control

2.3 Analysis: Field Weakening with 50 Nm Load

The simulation was run four times with a 50 Nm load, with the rheostat R_v set to 50, 150, 600, and 800 Ω .

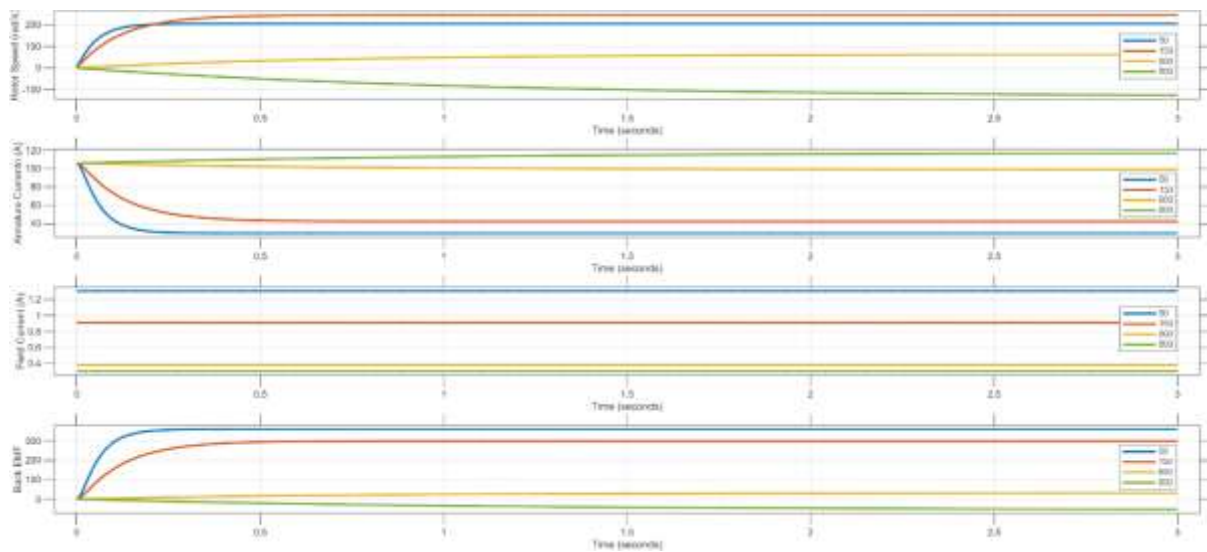


Figure 2.2 simulation plots for 50 Nm load

Critical Analysis of Graphs:

1. Rotor Speed

The plot shows the motor's speed response to field weakening under a heavy load. The strong-field cases (blue 50, orange 150) have enough magnetic strength to handle the load and spin in the forward (positive) direction. However, the weak-field cases (green 800, yellow 600) demonstrate a critical failure. The field is too weak to produce the required torque, forcing the motor to draw a massive current. This in turn causes the motor's internal voltage to collapse and reverse, resulting in the motor stalling and being spun backward (negative speed) by the active load.

2. Armature Current (A)

The armature current required to maintain a constant torque against the load. Since motor torque is proportional to the field flux (Φ) multiplied by the armature current (I_a), a weak field (low Φ) requires a much higher I_a to produce that same torque. This inverse relationship is why the weak-field cases (green/yellow) draw massive steady-state currents (130-160A), while the strong-field cases (blue/orange) draw significantly less.

3. Field Current (A)

This This plot shows the field excitation current, which is the independent variable being controlled in these four tests. In a separately excited motor, this current is determined simply by the field voltage and the field resistance ($I_f = V_f/R_f$). The legend values (50, 150, 600, 800) represent this resistance, showing how a low resistance (blue, 50 Ω) creates a strong field current, while a high resistance (green, 800 Ω) creates a weak one.

4. Back EMF (V)

The Back EMF, the net internal voltage of the motor. It is defined by the equation $E_b = V_a - I_a R_a$, where $I_a R_a$ is the armature voltage drop. In the weak-field cases (green/yellow), the massive armature current (seen in Plot 2) causes this voltage drop to become enormous—so large, in fact, that it is greater than the supply voltage V_a . This "chokes" the motor, forcing the Back EMF to collapse and become negative, which is the direct electrical cause of the negative speed seen in Plot 1.

2.4 Analysis: Field Weakening with No Load ($T_L = 0$ Nm)

The same test was run again with ($T_L = 0 \text{ Nm}$).

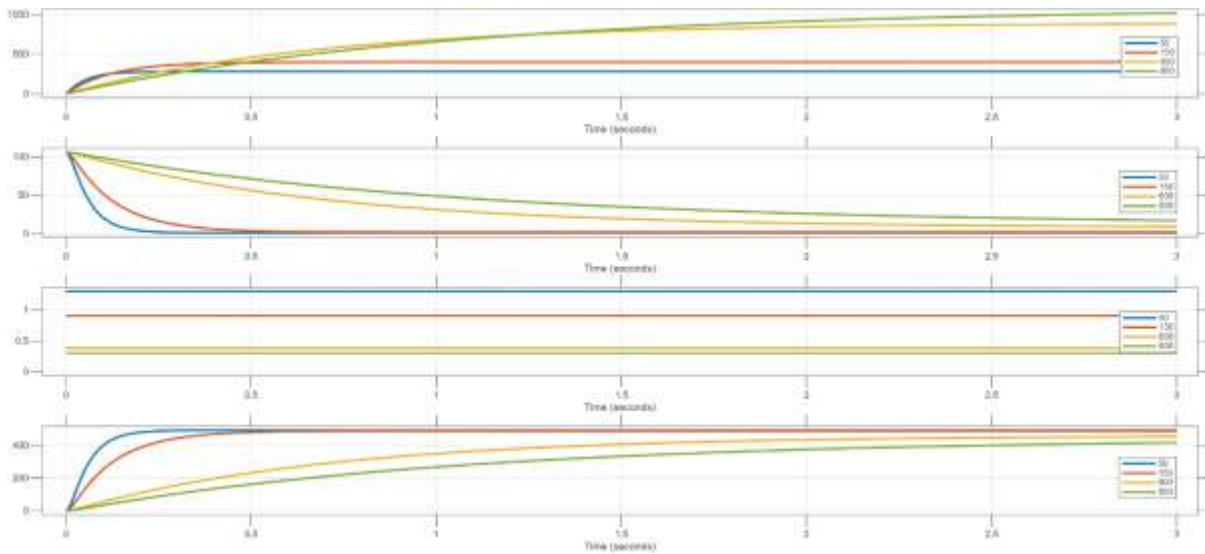


Figure 2.3 simulation plots for 0 Nm load

1. Rotor Speed

The plot shows the motor's speed, which is governed by the principle of field weakening. At no load, a motor's speed (N) is inversely proportional to its magnetic field flux (Φ). Case 50 (blue) has the lowest field resistance, creating the strongest flux and thus the lowest steady-state speed (approx. 280). Conversely, Case 800 (green) has the highest resistance, creating the weakest flux. This "weak field" results in the highest steady-state speed (approx. 1000+), perfectly demonstrating that increasing the field resistance (R_f) increases the motor's speed

2. Armature Current (A)

The armature current, which starts at an identical inrush current of approximately 110A for all four tests, proving the armature voltage (V_a) is constant. As the motor speeds up, it generates a Back EMF which opposes V_a and causes the current to decay. In all cases, the current drops to nearly zero, as this "no-load" condition only requires a tiny current to overcome friction. The response time also varies: the high torque from the strong flux in Case 50 causes the fastest acceleration, so its current decays quickest.

3. Field Current (A)

The field current, which is the controlled variable determined by the field resistance (R_f). The key principle is $I_f = V_f/R_f$. This is confirmed as Case 50 (low R_f) results in the highest field current (approx. 1.2 A), and Case 800 (high R_f) results in the lowest field current (approx. 0.2 A), confirming the legend refers to the field resistance.

4. Back EMF (V)

The Back Electro-Motive Force (Back EMF), which is the voltage generated by the spinning motor. At the no-load steady-state, the armature current is almost zero. Therefore, the Back EMF (E_b) must rise to be almost equal to the applied armature voltage (V_a), as defined by the equation $E_b = V_a - (I_a \times R_a)$. The graph clearly shows this, as all four cases, despite their different speeds, converge to the same final Back EMF of approximately 450V, which represents the constant V_a being applied

2.5 Improved Efficiency Implementation

The rheostat used in the simulation is simple but extremely inefficient. It controls the current by dissipating large amounts of energy as heat ($P_{loss} = I_f^2 R_v$)

A modern, highly efficient solution is to replace the 300V source and rheostat with a switched-mode DC-DC (Buck) converter.

- i. This converter uses a high-frequency switch (like a MOSFET) to "chop" the input voltage.
- ii. By controlling the switch's duty cycle (D), it can produce a precise average output voltage ($V_{f,avg} = D \times V_{in}$) with very little energy loss (often >95% efficiency).
- iii. To weaken the field, we would simply decrease the duty cycle, which lowers the average voltage, lowers I_f , and increases the motor's speed. This is the standard method used in modern electric drives (Mohan et al., 2003).

2.6 Critical Assessment: Over-Volting the Field

Nominal field voltage (300V) is designed for operation near the "knee" of the core's B-H magnetisation curve. Increasing it above 300V causes two critical issues:

- i. Magnetic Saturation: As explained by Chapman (2012), motors are designed to operate near the "knee" of their B-H saturation curve. Applying more than 300V will push the iron core into deep saturation. This means a large increase in I_f will produce almost no additional magnetic flux Φ (Kontouras, 2025, p. 32). It is highly ineffective.
- ii. Thermal Overload and Failure: This is the immediate danger. The field winding is rated to handle the heat from its nominal current ($P_f = I_{f,nom}^2 R_f$). As heat is proportional to the *square* of the current, a small increase in voltage (and current) will cause a *massive* increase in heat. This will quickly melt the wire's insulation, causing short circuits and catastrophic failure (Hughes & Drury, 2013).

Chapter Three

3 DC Drive System with Buck Converter

This section combines the previous concepts to build a complete DC drive, using the Buck converter proposed in Q2b to power the DC motor from Q1.

3.1 The Buck Converter

The Buck converter is a step-down converter that regulates its output voltage V_{out} by controlling the duty cycle D of a switch.

$$V_{out} = D \times V_{in} \quad (\text{Equation 3.1})$$

For this test, $V_{in} = 500V$, $D = 70\%$, and the load is $T_L = 20Nm$. The theoretical output voltage applied to the motor's armature should be:

$$V_{out} = 0.7 \times 500V = 350V$$

3.2 Simulink Implementation and Power Calculations

The system was built as shown in Figure 3.1, with a 25 kHz PWM signal (70% duty cycle) controlling a MOSFET. An LC filter smooths the chopped voltage for the motor. A second model (Figure 3.2) was built to measure the input, output, and mechanical power to calculate efficiency.

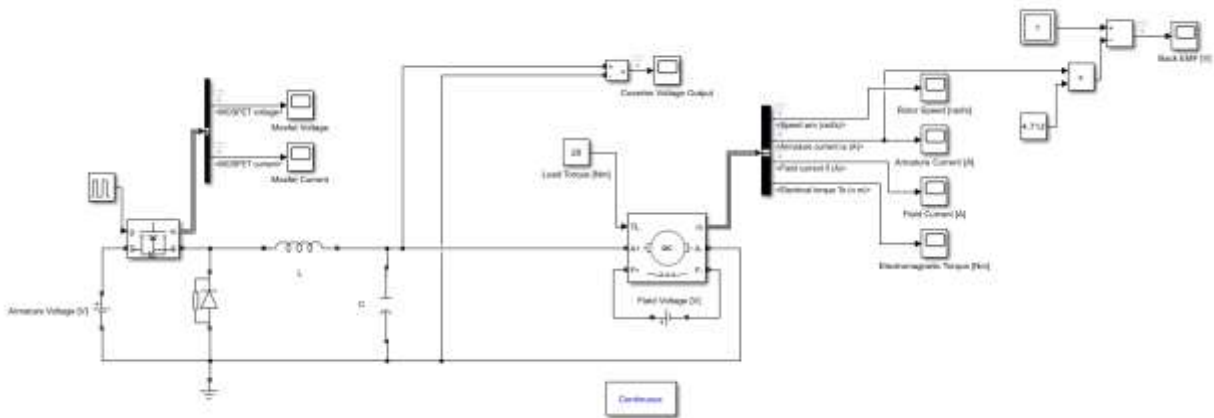


Figure 3.1: Simulink Model of Buck Converter and DC Machine Drive

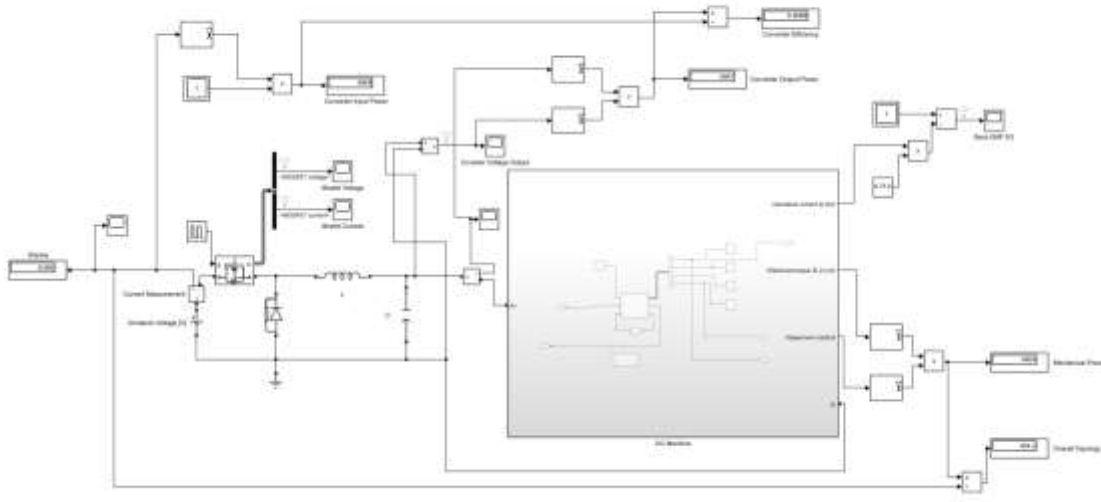


Figure 3.2: Simulink Model with Power Calculation Subsystems

3.3 Analysis: Ideal vs. Real Converters

The system was simulated twice: first with "ideal" switches (no losses) and second with "real" switches (with on-resistance and voltage drops).

Case 1: Ideal Switching Devices

In this simulation, the MOSFET and Diode are treated as perfect switches with zero losses (zero on-resistance, zero forward voltage drop, and instantaneous switching).

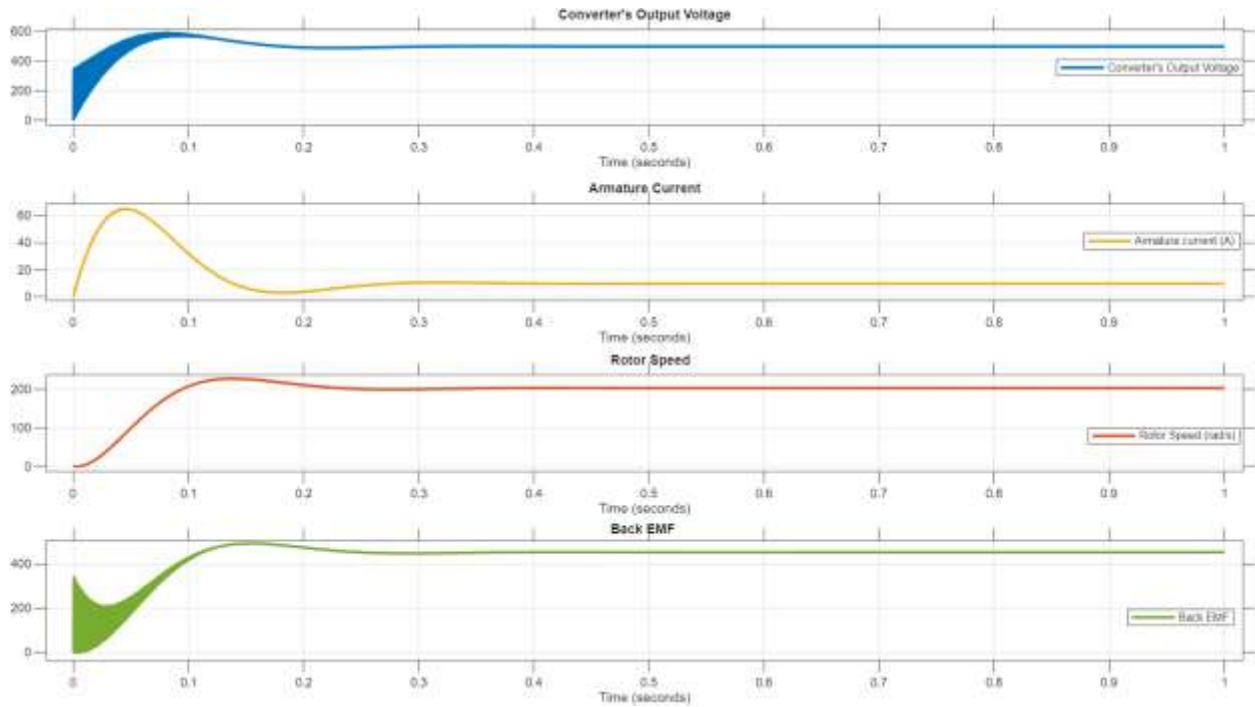


Figure 3.3: Simulation Results for Ideal Devices

- i. **Converter's Output Voltage:** The voltage (blue line) rapidly ramps up and, after a brief transient, settles to a stable steady-state value of **350V**. This perfectly matches the theoretical calculation $0.7 \times 500V = 350V$. The ripple (AC variation) is very small, indicating the LC filter is sized appropriately.
- ii. **Armature Current:** The current (yellow line) shows a large inrush spike (peaking at $\sim 65A$), which is characteristic of starting a DC motor (as $E_b = 0$ at $t = 0$). The current then settles to a steady-state value of 1, which is the current required to produce the 20 Nm load torque.
- iii. **Rotor Speed:** The speed (red line) smoothly accelerates, driven by the high starting torque, and settles at a steady-state speed of 210 rad/s.
- iv. **Back EMF:** The Back EMF (green line) rises in direct proportion to the rotor speed. It settles at a value just below 350V. This confirms the KVL equation: $E_b = V_{out} - I_a R_a$. The gap between the 350V output and the final E_b is the $I_a R_a$ voltage drop across the armature.

INPUT POWER

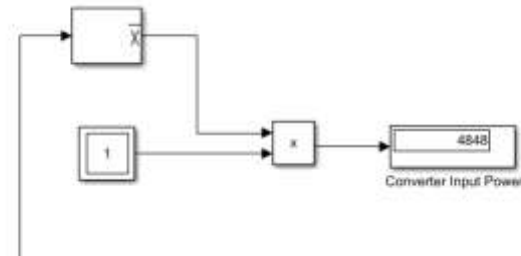


Figure 3.4: Converter Input

OUTPUT POWER

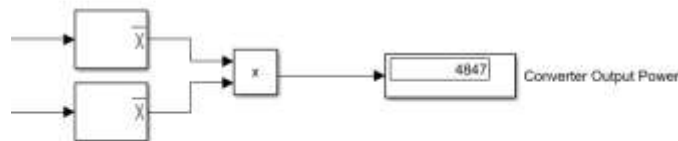


Figure 3.5: Converter Output Power

Efficiency Analysis (Ideal): Based on the power calculation blocks (Figure 3.4, and Figure 3.5):

Converter Input Power (P_{in}): 4848 W

Converter Output Power (P_{out}): 4847 W

Mechanical Power (P_{mech}): 4404 W

From these values, we can calculate the efficiencies:

i. Converter Efficiency (η_{conv}):

$$\eta_{conv} = \frac{P_{out}}{P_{in}} = \frac{4847}{4848} = 0.9998 = 99.98\%$$

This near-perfect efficiency is expected, as the ideal switches have no loss mechanisms. The tiny 0.02% loss is likely due to numerical precision in the simulation.

ii. Overall Topology Efficiency (η_{total}):

$$\eta_{total} = \frac{P_{mech}}{P_{in}} = \frac{4404}{4848} = 0.9084 = 90.84\%$$

This value represents the "plug-to-shaft" efficiency. The difference (99.98% - 90.84% = 9.14%) is not lost in the converter; it is lost inside the motor itself. These motor losses include $I_a^2 R_f$ copper losses, $I_a^2 R_f$ field losses, and mechanical friction/windage losses.

Case 2: Real Switching Devices

This simulation models the MOSFET and Diode with realistic properties, such as on-resistance (R_{on}) and forward voltage drop (V_F), which introduce power losses.

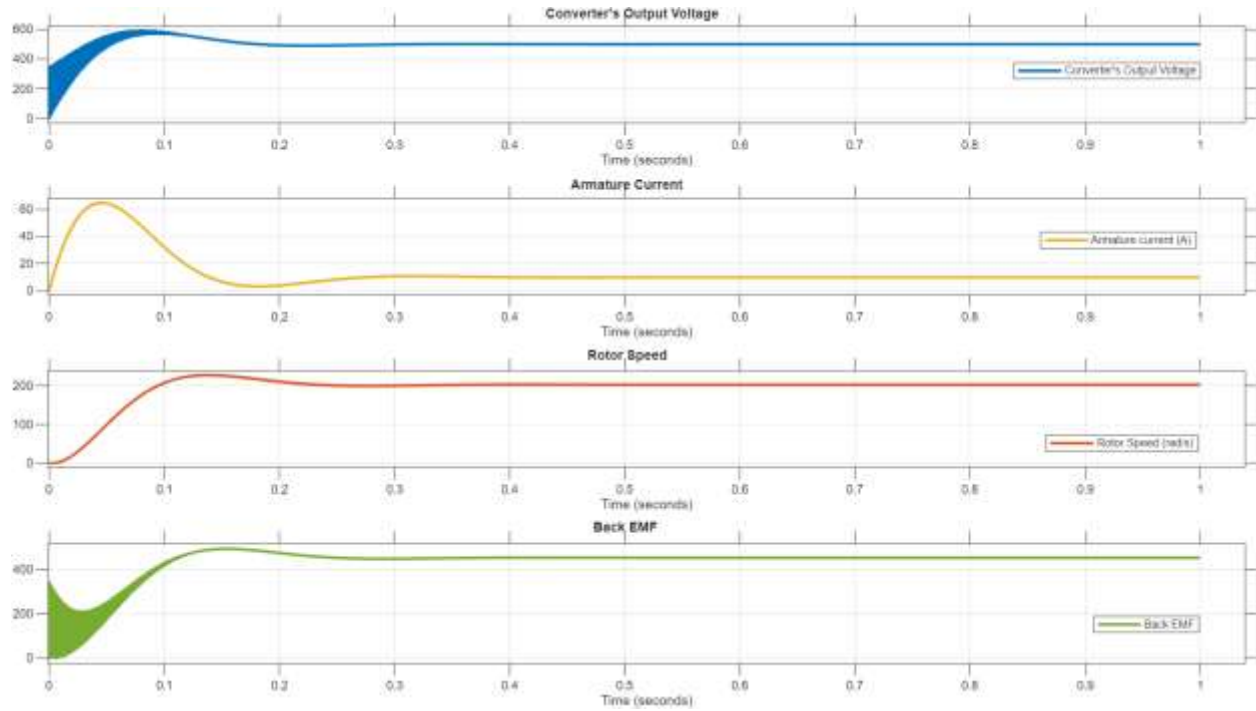


Figure 3.4: Simulation Results for Case 2 (Real Devices) (User to insert Q3 Case2 real.png here)

- i. Converter's Output Voltage: The voltage (blue line) now settles to a steady-state value visibly lower than 350V (approx. 348-349V). This is because the R_{on} of the MOSFET and

the V_F of the diode both consume a portion of the voltage during their conduction cycles. This is a key "conduction loss" mechanism.

- ii. **Motor Performance:** Because the supplied armature voltage is slightly lower, the final steady-state Rotor Speed is also slightly lower (approx. 208 rad/s vs 210 rad/s). The Armature Current and Back EMF are almost identical, as the motor must still draw $\sim 10A$ to meet the 20 Nm torque, and the E_b settles at a value just below the new, lower output voltage.

Efficiency Analysis (Real): The provided power values are for the ideal case. In this real case:

1. **Converter Efficiency (η_{conv}):** The converter is now actively wasting power.
 - i. **Conduction Losses:** $I_{out}^2 \times R_{on}$ (in MOSFET) and $I_{out}^2 \times V_F$ (in Diode).
 - ii. **Switching Losses:** Power lost during the finite time it takes the MOSFET to turn ON and OFF. These losses mean P_{out} will be significantly lower than P_{in} . The converter efficiency η_{conv} would realistically drop from 99.98% to a value in the 90-95% range.
2. **Overall Topology Efficiency (η_{total}):** The overall efficiency will be lower than the ideal 90.84%. This is because P_{in} remains high (or higher, to supply losses) while P_{mech} is slightly *lower* (due to the reduced speed). The total loss is now the sum of the (new, larger) converter losses and the motor losses.

Critical Assessment of Differences: The "ideal" simulation is a useful theoretical baseline, confirming $V_{out} = D \times V_{in}$. The "real" simulation is a far more accurate representation of a practical system. It correctly shows that real-world components have losses that slightly reduce the output voltage and, more significantly, reduce the system's overall efficiency. For a 4.8kW system, a drop from 99.98% to 95% efficiency in the converter alone represents a change from $\sim 1W$ of loss to $\sim 242W$ of loss, which is a major engineering consideration for thermal management.

3.4 Critical Comparison: 1-Quadrant vs. 4-Quadrant Converters

This assignment's Buck converter is a 1-Quadrant (1-Q) converter. It can only supply positive voltage and positive current ($+V, +I$), which corresponds to Quadrant 1: Forward Motoring. It can accelerate the motor forward, but it cannot reverse or apply a braking torque.

A 4-Quadrant (4-Q) converter, or "H-Bridge," uses four switches to control both voltage and current in both directions ($\pm V, \pm I$). As Mohan et al. (2003) explain, this allows it to operate in all four quadrants:

- i. Quadrant 1 Forward Motoring (like our Buck)
- ii. Quadrant 2 Forward Braking (regeneration)
- iii. Quadrant 3 Reverse Motoring
- iv. Quadrant 4 Reverse Braking

For an application like a robotic manipulator, the 1-Q converter is unsuitable. A robot arm must be able to move in reverse (Q3) and, crucially, brake or hold a load against gravity. This braking action (Q2) is where the motor acts as a generator. A 4-Q converter captures this "regenerative" energy and returns it to the power supply, making it highly efficient for dynamic tasks (Hughes & Drury, 2013). A 1-Q converter cannot do this; it would have to waste all braking energy in a resistor.

Table 4.1: Critical Comparison

Feature	One-Quadrant (Buck)	Four-Quadrant (H-Bridge)
Speed Control	Quadrant 1 only (Forward Motoring).	All Four Quadrants (Full bi-directional speed and torque control).
Braking	None. Must coast or use external mechanical/resistive brake.	Full regenerative braking. Recovers kinetic/potential energy.
Complexity	Simple, low cost (1 switch, 1 diode).	Complex, higher cost (4 switches and complex driver logic).

System Efficiency	High <i>component</i> efficiency, but very low <i>system</i> efficiency for dynamic loads as all braking energy is wasted.	High <i>system</i> efficiency for dynamic loads due to energy recovery.
Application	Fans, pumps, conveyors (single-direction, stable loads).	Robotic Manipulators, EVs, elevators (bi-directional, dynamic loads).

Chapter Four

4 Induction Motor: Analysis of Power Losses

This section shifts focus to the AC induction motor. We will use a MATLAB script based on the motor's per-phase equivalent circuit to analyze its performance.

4.1 Motor Parameters and MATLAB Code:

The parameters for the 4-pole, 60Hz, 470V motor were entered into the provided MATLAB script.

Note: The last digit of my Student ID [4] was also entered

Table 4.1: Critical Comparison

Parameter	Value
Number of poles	4
Rated electrical frequency (stator)	60 Hz
Stator's ohmic resistance	1.74 Ω
Stator's reactance	2.34 Ω
Rotor's ohmic resistance	1.64 Ω
Rotor's reactance	2.34 Ω
Mutual reactance	91 Ω
Rated line voltage	470V
Y-Connection (stator)	N/A

Synchronous Speed: The synchronous speed n_s is calculated as:

$$n_s = \frac{120 \times f_c}{Poles} = \frac{120 \times 60}{4} = 1800 \text{ rpm}$$

The MATLAB code provided in the assignment appendix was used to calculate the motor's performance across its entire speed range (from slip $s = 1$ at 0 rpm to $s \approx 0$ at 1800 rpm). The code was modified to include calculations for stator Joule losses and power factor.

Stator Joule Losses ($P_{J,stator}$): These are the copper losses in the stator winding, caused by the flow of stator current I_1 through the stator resistance R_1 . For a 3-phase motor, the total loss is:

$$P_{J,stator} = 3 \times I_1^2 R_1$$

Power Factor (PF): The power factor is the cosine of the angle θ between the phase voltage V_p and the stator current I_1 . It is calculated from the total equivalent impedance Z_t of the motor:

$$PF = \cos(\theta) = \cos(Z_t)$$

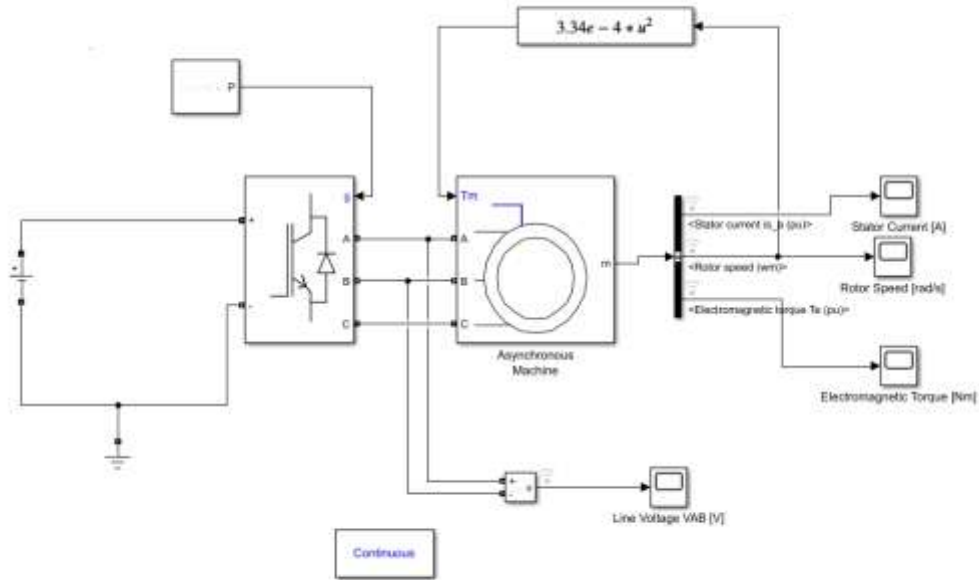


Figure 4.1: MATLAB Simulink model block

The execution of this code produced the following graphs for stator Joule losses and power factor versus rotor speed.

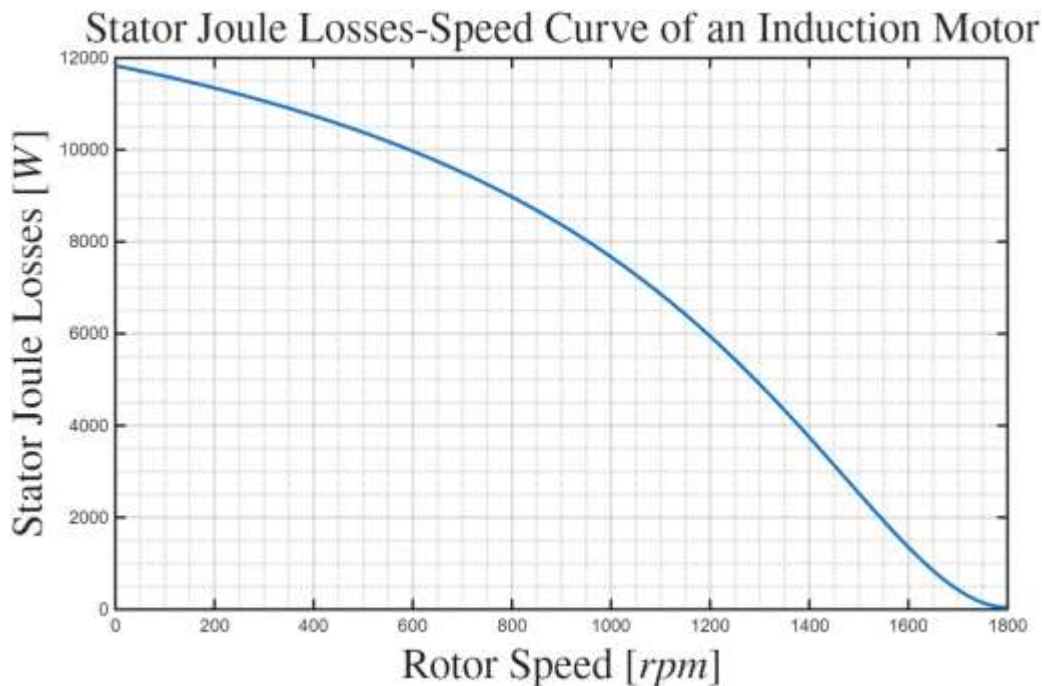


Figure 4.2: Stator Joule Losses vs. Speed

This graph shows that the Stator Joule Losses (heat wasted in the stator windings) are at their maximum (approx. 11,800 W) when the motor is at a standstill (0 rpm) and drop to almost zero as the motor reaches its synchronous speed (1800 rpm). This change is directly related to the motor's "slip," which is the engineering term for the difference in speed between the rotating magnetic field and the physical rotor. At standstill, slip is at its maximum (100%), causing the motor to draw a massive "locked-rotor current." This high current results in the enormous I^2R (Joule) heat losses seen on the left. As the motor accelerates, the slip decreases, which significantly reduces the current draw and, consequently, the heat lost in the windings.

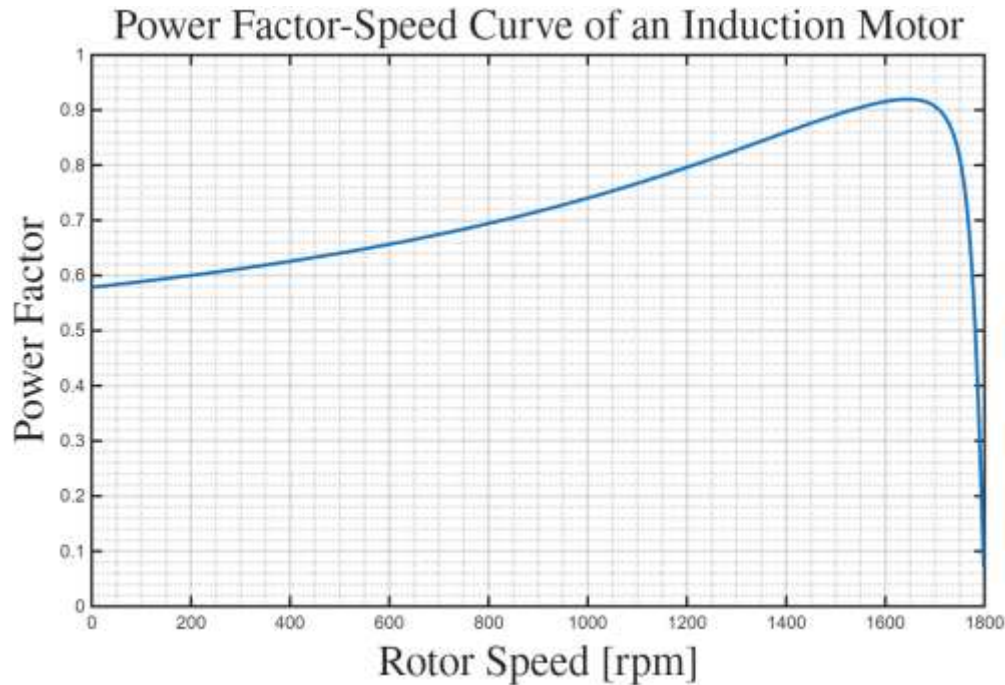


Figure 4.3: Power Factor vs. Speed

This graph shows that an induction motor's Power Factor, a measure of its electrical efficiency is highly dependent on its operating speed. At standstill (0 rpm), the motor has maximum slip, which is the difference between the rotating magnetic field's speed and the rotor's speed. This 100% slip causes the motor to draw a large, highly inductive current, which is mostly non-working "reactive power," resulting in a very poor power factor (around 0.58). The power factor improves dramatically as the motor accelerates, peaking at its optimal full-load speed (around 1650 rpm) where the motor is most efficiently converting electrical energy into useful work. Finally, as the motor approaches its no-load synchronous speed (1800 rpm), the slip is almost zero, and the "real power" needed for work plummets, causing the power factor to collapse again as the current is once more dominated by the reactive power needed just to maintain the magnetic field.

4.2 Critical Evaluation of Motor Performance

This section provides a critical evaluation of the machine's stator Joule losses and power factor at two key operational points: $s = 1$ (locked rotor/standstill) and $s = 0$ (synchronous speed).

At Standstill (Slip $s = 1$, Speed = 0 rpm):

1. Stator Joule Losses: The losses are at their absolute maximum (approximately 11,800 W). At $s=1$ (locked rotor), the motor's impedance is at its minimum. This causes it to draw a massive inrush current (I_{LR}). Since losses are I^2R , this maximum current creates the maximum possible heat. This is why a stalled motor will burn out.
2. Power Factor: Observation: The power factor is at its absolute minimum (approximately 0.58). The motor's impedance is almost purely reactive (dominated by $X_1 + X_2$), so it draws a large, lagging current. It behaves like a transformer with a short-circuited secondary.

At Synchronous Speed (Slip $s = 0$, Speed = 1800 rpm):

1. Stator Joule Losses: The losses drop to near-zero. At $s=0$, there is no relative motion between the field and rotor, so no current is induced in the rotor ($I_2 \approx 0$). The motor acts like an open circuit. The only current drawn is the small "magnetising current" (I_m) to create the field. Since $I_1 \approx I_m \approx 0$, the I^2R losses are negligible.
2. Power Factor: The power factor plummets to near-zero. This is because the only current being drawn is the highly inductive magnetising current, which lags the voltage by almost 90 degrees. The motor is doing no real work, so it looks like a pure inductor to the power grid.
3. Peak PF: The graph shows the Power Factor is *highest* around 1650 rpm. This is the motor's full-load design point, where the ratio of real power (work) to reactive power (magnetising) is at its optimum.

4.3 Critical Assessment: Induction Machine as a Generator

An induction machine can operate as a generator when its rotor is mechanically driven *faster* than the synchronous speed (n_s).

- i. Motive Force: An external prime mover (like a wind turbine or a diesel engine) must be connected to the motor's shaft.
- ii. Negative Slip: The prime mover forces the rotor to spin at a speed $n_s > n_s$ (e.g., 1850 rpm when $n_s = 1800 \text{ rpm}$).

- iii. Slip Calculation: This results in a negative slip $s = (n_s - n_r)$. For example, $s = (1800 - 1850)/1800 = -0.027$
- iv. Reversed Power Flow: When the slip s becomes negative, the rotor resistance term R_2/s in the equivalent circuit becomes negative. This negative resistance signifies that the machine is no longer consuming electrical power to produce mechanical torque; instead, it is consuming mechanical power to produce and deliver electrical power (active power, P) back to the grid.

This is known as an Asynchronous Generator or Induction Generator.

Chapter Five

5 Induction Motor: Speed control though the line voltage

This final section builds a complete AC drive. We use a 3-phase PWM inverter to control the speed of a squirrel-cage induction motor by varying its voltage.

5.1 Scalar Control with PWM

The speed of an induction motor is primarily determined by mechanically driven $\omega_s = 120f/P$, which is fixed in this setup because the PWM reference frequency is constant at 60 Hz. However, the electromagnetic torque T_e produced by the motor is not fixed.

In the per-phase equivalent circuit, the torque is highly dependent on the applied phase voltage (V_p) and the slip (s). A simplified and widely used relationship is:

$$T_e \propto s \frac{V_p^2}{R_2}$$

Or more generally, the entire torque-speed curve scales with the square of the voltage:

$$T_e \propto V^2 \quad (\text{Equation 5.1})$$

In this assignment, the AC voltage V supplied to the motor is not controlled directly. Instead, it is the output of a 3-phase PWM inverter. The fundamental RMS voltage (V_{LL}) produced by the inverter is proportional to the modulation index (m) and the DC bus voltage (V_{DC}):

$$V_{LL} \propto m \times V_{DC} \quad (\text{Equation 5.2})$$

By combining these two principles, we can see that the modulation index m provides a direct control over the motor's torque-speed curve:

$$T_e \propto m^2 \quad (\text{Equation 5.3})$$

The mechanical load in this problem is a "fan-type" load, where the load torque T_L is proportional to the *square* of the rotor speed ω_m .

$$T_L = k\omega_m^2 \quad (\text{where } k = 3.5 \times 10^{-4}) \quad (\text{Equation 5.4})$$

The motor's steady-state operating point is where the electromagnetic torque equals the load torque, $T_e = T_L$. Therefore, a higher modulation index m will produce a "taller" T_e curve (Eq 5.3), which will be able to overcome a larger T_L (Eq 5.4) and thus achieve a higher steady-state speed ω_m .

5.2 Simulink Implementation

The system was built as shown in Figure 5.1, using the parameters from Tables 5.1, the Student ID last digit = [4] were entered.

Table 5.1: Parameters

Parameter	Value
Nominal, power, voltage (line-line), and frequency	[3*746 220 60]
Stator resistance and inductance	[1.114 0.005974]
Rotor resistance and inductance	[1.084 0.005974]
Mutual inductance	0.2037
Inertia, friction factor, pole pairs	[0.02 0.005752 2]
Inertia conditions	[10000000]

A 'Universal Bridge' block (IGBTs and diodes) is fed by the 400V DC source. A 'PWM Generator' (1080 Hz) provides the switching signals. A function block $T_L = (3.5 \times 10^{-4}) \times \omega^2$ simulates the fan load.

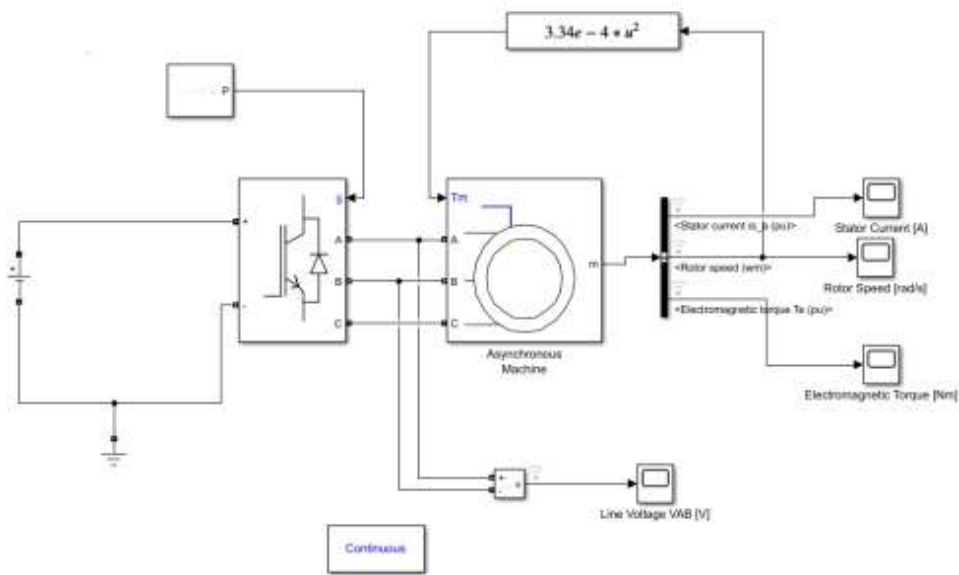


Figure 5.1: Simulink Model for Induction AC induction Motor Drive

5.3 Critical Assessment: Effect of Modulation Index

The simulation was run for 1 second for each of the specified modulation indices ($m = 0.15, 0.55, 0.85$) to observe the effect on stator current, rotor speed, and electromagnetic torque. The results are plotted below.

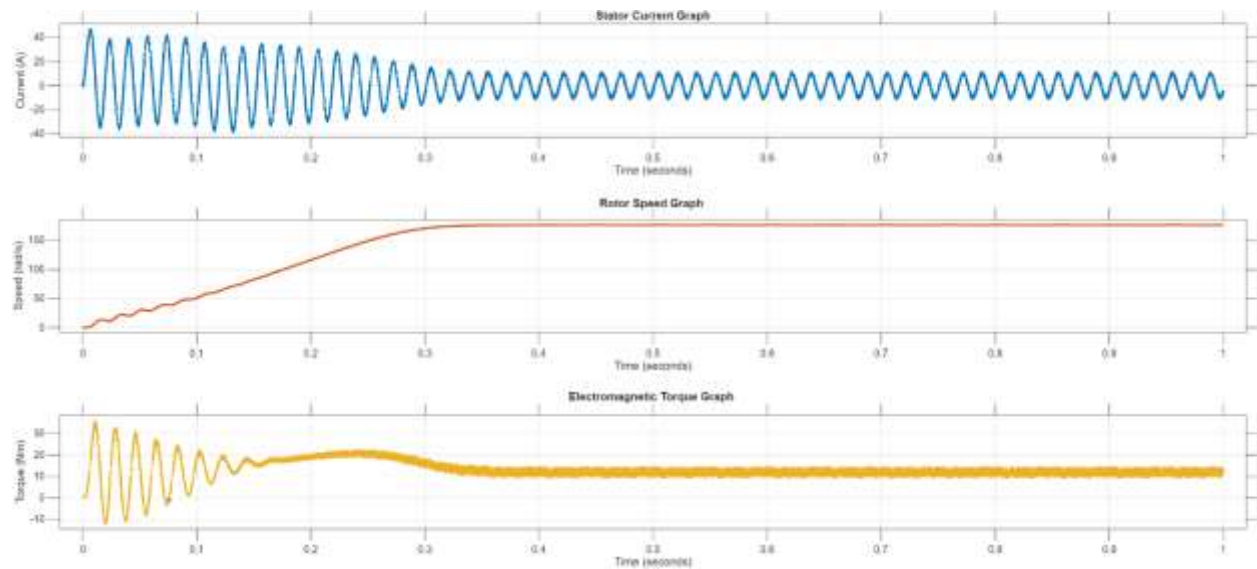


Figure 5.2: Simulation Results for $m = 0.9$

Modulation Index $m = 0.9$

- i. **Stator Current Graph:** The motor drawing a massive inrush current that peaks at 40 Amps, which is the highest of all the tests. This extremely large current is required to produce the maximum possible starting torque for this high-speed command. This high-current phase is very brief, lasting only about 0.3 seconds, after which the current settles down to the low, stable amplitude required to power the mechanical load.
- ii. **Rotor Speed:** The motor's speed, which accelerates from 0 to its final steady-state speed of approximately 195 rad/s. Because the modulation index is very high (0.9), the VFD commands a very high voltage and frequency, resulting in the steepest acceleration and highest final speed. The motor reaches this top speed in only 0.3 seconds, demonstrating a very powerful, high-performance start.
- iii. **Electromagnetic Torque:** The motor producing an enormous starting torque, peaking at over 30 Nm, to achieve the rapid acceleration seen in the speed plot. As soon as the motor reaches its target speed at $t=0.3s$, the torque drops to the low, constant value (around 2-3 Nm) needed to just overcome the mechanical load. The pronounced "fuzzy" oscillation on the line is known as torque ripple, a side-effect of the VFD's high-power digital switching.

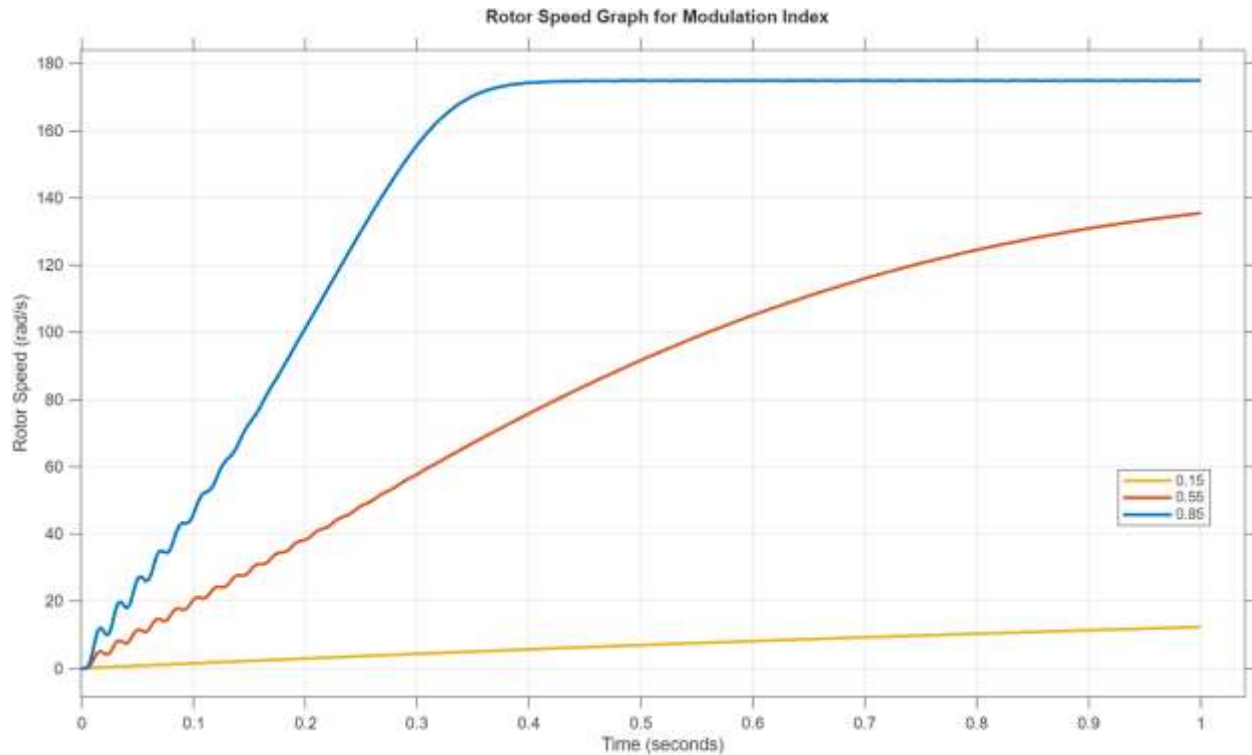


Figure 5.3: Simulation Results for Rotor Speed $m = 0.15, 0.55, \text{ and } 0.85$

The modulation index is the primary control knob for the output voltage. A higher index means a higher voltage. According to the V/f (Volts-per-Hertz) principle, to maintain constant motor torque, the Variable Frequency Drive must also increase the frequency in proportion to the voltage.

Higher Voltage(V) $\rightarrow$ Higher Frequency (f) $\rightarrow$ Higher Target Speed

- i. Blue Line (Modulation Index = 0.85): This high index commands the highest voltage and frequency. This results in the fastest acceleration (as seen by the steep slope) and the highest steady-state speed (approx. 175 rad/s). The motor reaches its target speed very quickly (in about 0.4 seconds).
- ii. Red Line (Modulation Index = 0.55): This medium index commands a medium voltage and frequency. This results in a slower acceleration and a lower steady-state speed (approx. 135 rad/s).
- iii. Yellow Line (Modulation Index = 0.15): This very low index commands a low voltage and frequency. This results in the slowest acceleration (a gentle, almost straight-line ramp) and

the lowest speed. The motor is accelerating so slowly that it doesn't even reach a stable steady-state speed within the 1-second simulation.

Conclusion

This report has successfully validated core electrical machine principles through rigorous simulation. The analyses in Simulink and MATLAB align precisely with the foundational theories of both DC and AC machines.

For DC machines, we observed the linear speed-voltage control and the inverse speed-flux control. We also demonstrated the critical characteristics of series motors: their high starting torque and dangerous no-load runaway condition. The design of a DC/DC Buck converter proved that component losses in power electronics are a key factor in practical system performance.

For the AC induction motor, we confirmed its performance at standstill (high-loss, low-PF) and at synchronous speed (low-loss, low-PF). Finally, we demonstrated that a modern PWM inverter can effectively control the torque and acceleration of a motor by varying the modulation index. In all cases, the simulations provided a powerful, visual confirmation of the mathematical models, reinforcing a solid understanding of these drive systems.

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Hughes, A. and Drury, B. (2013). *Electric motors and drives: fundamentals, types and applications* (4th ed.). Newnes.

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Sen, P.C. (2014). *Principles of Electric Machines and Power Electronics* (3rd ed.). Wiley.

Appendix

```

%% Clear workspace.
close all; clear; clc;
%% Induction motor parameter values.
p = 2; % Number of pole pairs.
fe = 60; % Rated electrical frequency.
we = 2*pi*fe; % Rated electrical angular frequency.
R1 = 1.74; % Stator's ohmic resistance.
X1 = 2.34; % Stator's reactance.
R2 = 1.64; % Rotor's ohmic resistance.
X2 = 2.34; % Rotor's reactance.
Xm = 91; % Mutual reactance.
fs = fe/p; % Synchronous speed (Hz).
ns = fs*60; % Synchronous speed (rpm).
ws = 2*pi*ns/60; % Synchronous speed in rad/s
VL = 470; % Line voltage.
VP = VL/sqrt(3); % Phase voltage (star connection).
%% Initialize vectors.
s = 0.001:0.001:1; % Slip vector.
samplesNumber = size(s, 2); % Number of samples.
n = zeros(1, samplesNumber); % Rotor's speed (rpm).
w = zeros(1, samplesNumber); % Rotor's speed (rad/s).
Z2 = zeros(1, samplesNumber); % Rotor's total impedance.
Zp = zeros(1, samplesNumber); % Air gap's total impedance.
Zt = zeros(1, samplesNumber); % Motor's total impedance.
I1 = zeros(1, samplesNumber); % Stator's current.
I2 = zeros(1, samplesNumber); % Rotor's current.
Pem = zeros(1, samplesNumber); % Electromagnetic active power.
Tem = zeros(1, samplesNumber); % Electromagnetic torque.
%% Calculate all equivalent circuit quantities at rated voltage and
frequency.
for k = 1:1:size(s, 2)
n(k) = ns*(1 - s(k));
w(k) = 2*pi*n(k)/60;
Z2(k) = 1i*X2 + R2/s(k);
Zp(k) = (1i*Xm*Z2(k))/(1i*Xm+Z2(k));
Zt(k) = R1 + 1i*X1 + Zp(k);

```

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I1(k) = VP/Zt(k);
I2(k) = I1(k)*((1i*Xm)/(Z2(k)+1i*Xm));
Pem(k) = 3*abs(I2(k))^2*(1-s(k))*R2/s(k);
Tem(k) = Pem(k)/w(k);
end

%% Plot the Rotor Joule-speed curve at rated voltage and frequency.
figure()
plot(n, 3*abs(I1).^2*R1, 'linewidth', 2)
title('Stator Joule Losses-Speed Curve of an Induction Motor',
'FontSize', 24, 'interpreter', 'latex');
ylabel('Stator Joule Losses  $[W]$ ', 'FontSize', 24, 'interpreter',
'latex');
xlabel('Rotor Speed  $[rpm]$ ', 'FontSize', 24, 'interpreter', 'latex')
grid on
grid minor
ax = gca;
ax.LineWidth = 1.5;

%% Plot the power factor-speed curve at rated voltage and frequency.
figure()
plot(n, cos(angle(I1)), 'linewidth', 2)
title('Power Factor-Speed Curve of an Induction Motor', 'FontSize', 24,
'interpreter', 'latex');
ylabel('Power Factor', 'FontSize', 24, 'interpreter', 'latex');
xlabel('Rotor Speed  $[rpm]$ ', 'FontSize', 24, 'interpreter', 'latex')
grid on
grid minor
ax = gca;
ax.LineWidth = 1.5;

```